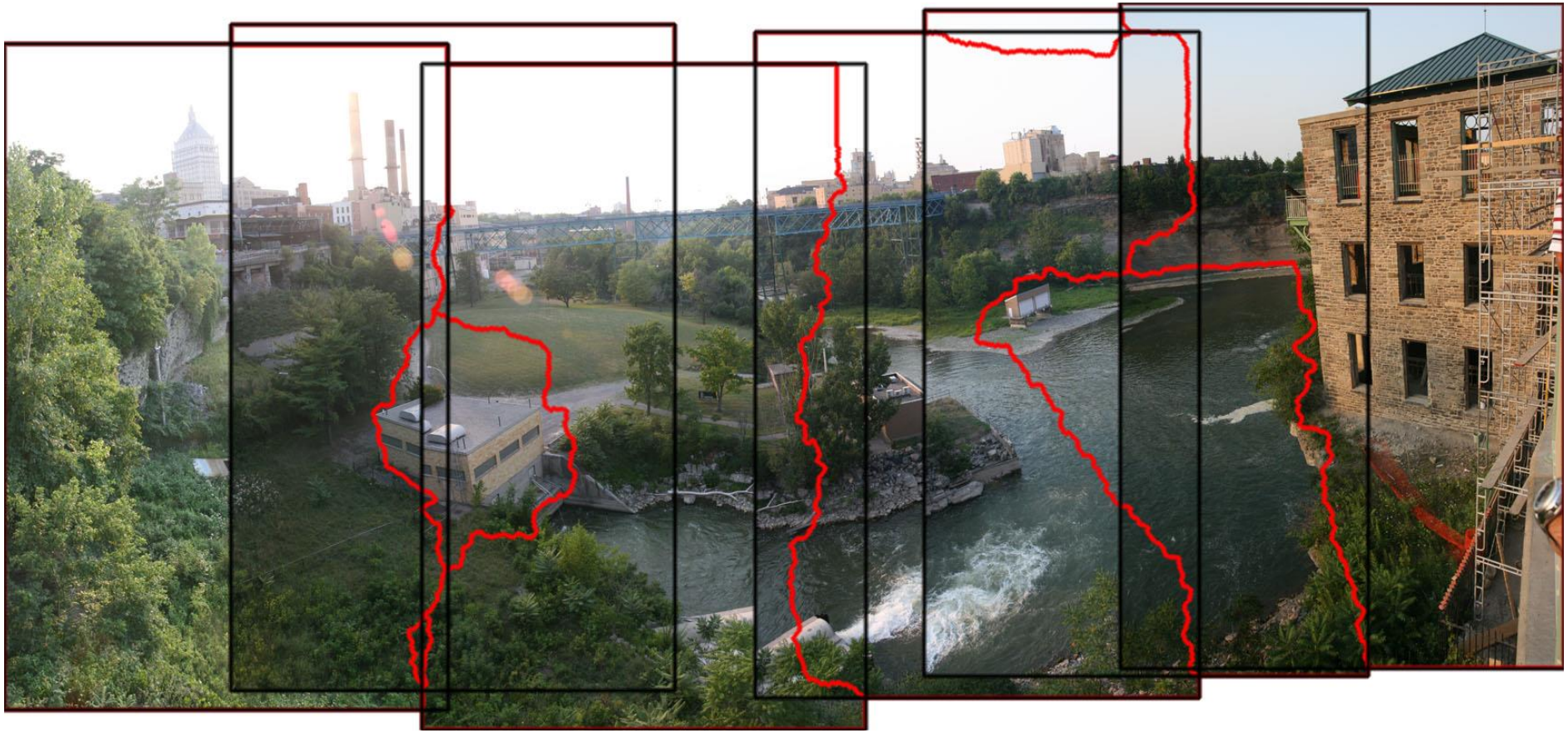
Image Alignment

Applications

- Image stitching
- Panorama
- Burst image denoising
- Handheld Multi-Frame super-resolution
- High Dynamic Range(HDR)
- 双摄/多摄
- Video Stabilization

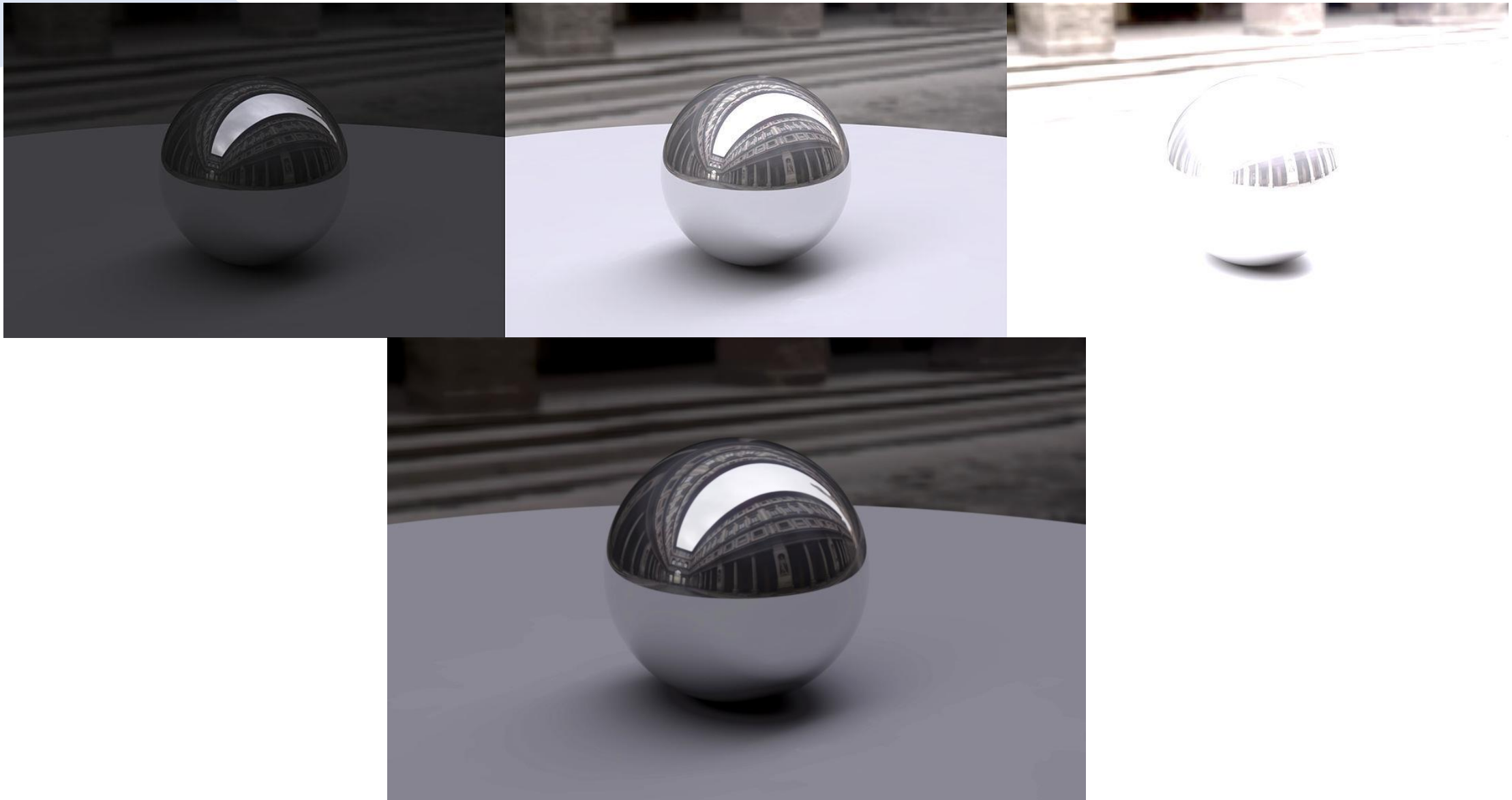
Image stitching
Panorama



Burst image denoising

Handheld Multi-Frame super-resolution

High Dynamic Range(HDR)



HYATT

HYATT

双摄/多摄

Video Stabilization

- 机械稳像(Mechanical Image Stabilization)
- 光学稳像(Optical Image Stabilization)
- 电子稳像(Digital Image Stabilization)

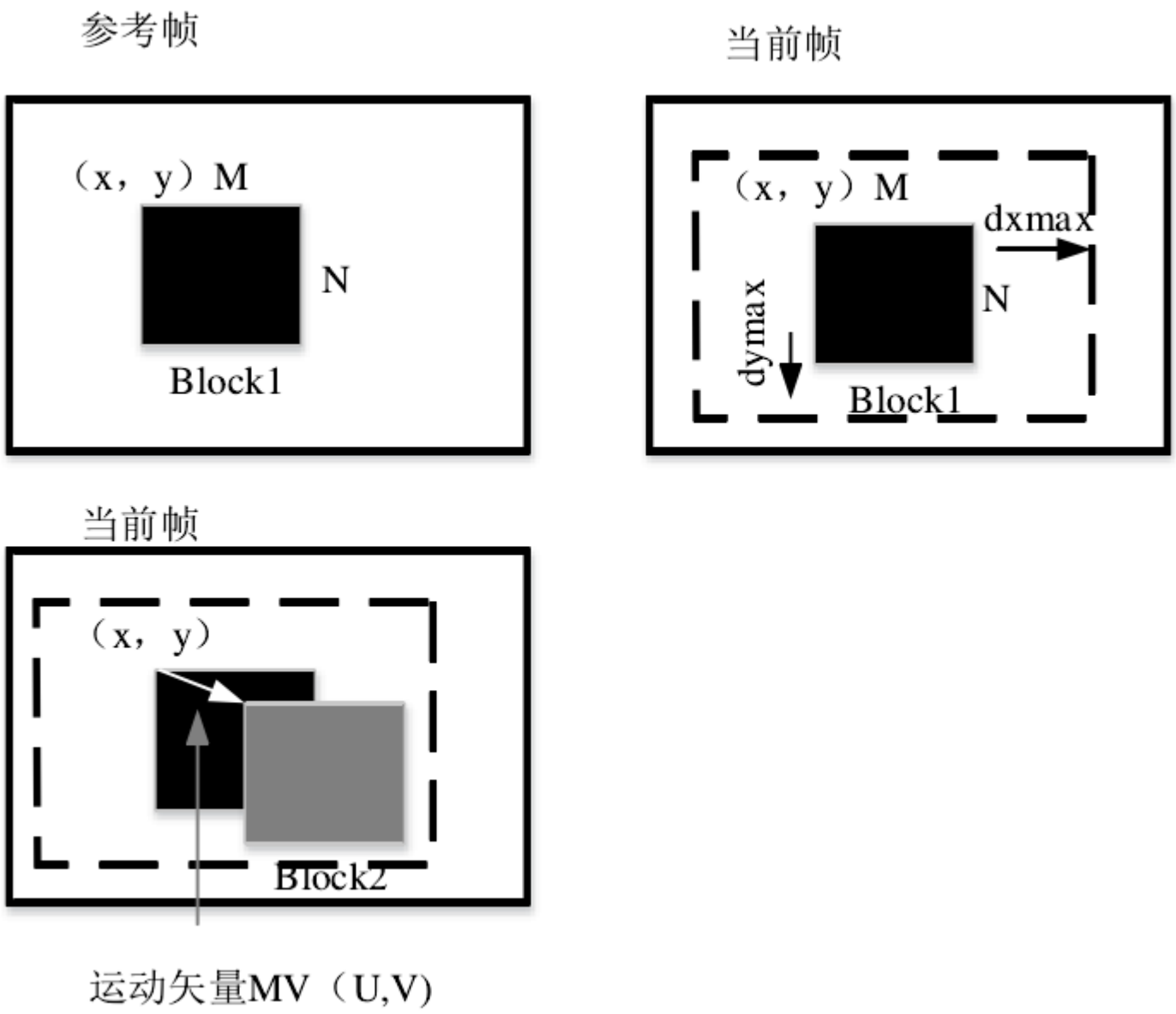
电子稳像(Digital Image Stabilization)

- 1.估计原始摄像机路径（运动估计）
- 2.计算新的平滑的摄像机路径（运动补偿）
- 3.基于平滑路径合成新的稳定的视频（图像修补）

1.估计原始摄像机路径（运动估计）

- 块匹配法
- 灰度投影法
- 相位相关算法
- 光流法
- 基于特征点的方法
- …
- 基于硬件：陀螺仪

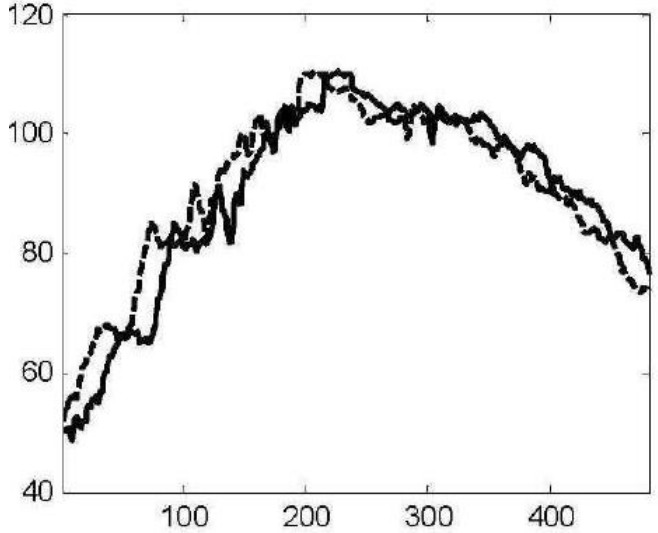
块匹配法


参考帧
（x，y）M
N
Block1
当前帧
（x，y）M
dxmax
dymax
N
Block1
当前帧
（x，y）
Block2
运动矢量MV（U,V)

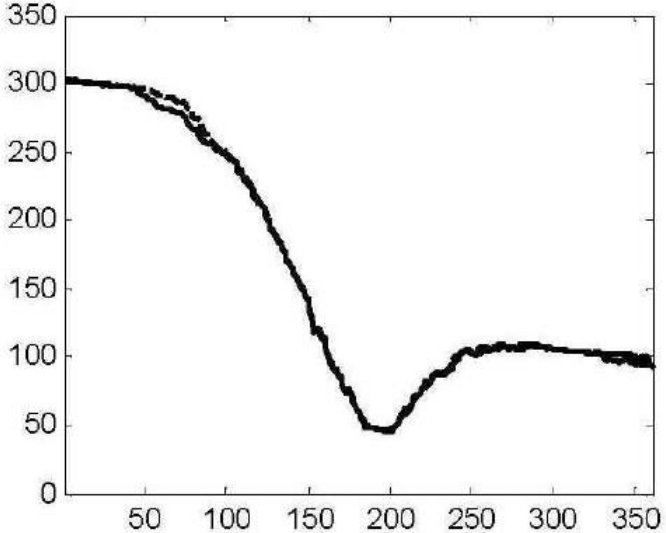
灰度投影法

（a）参考帧图像

（b）当前帧图像

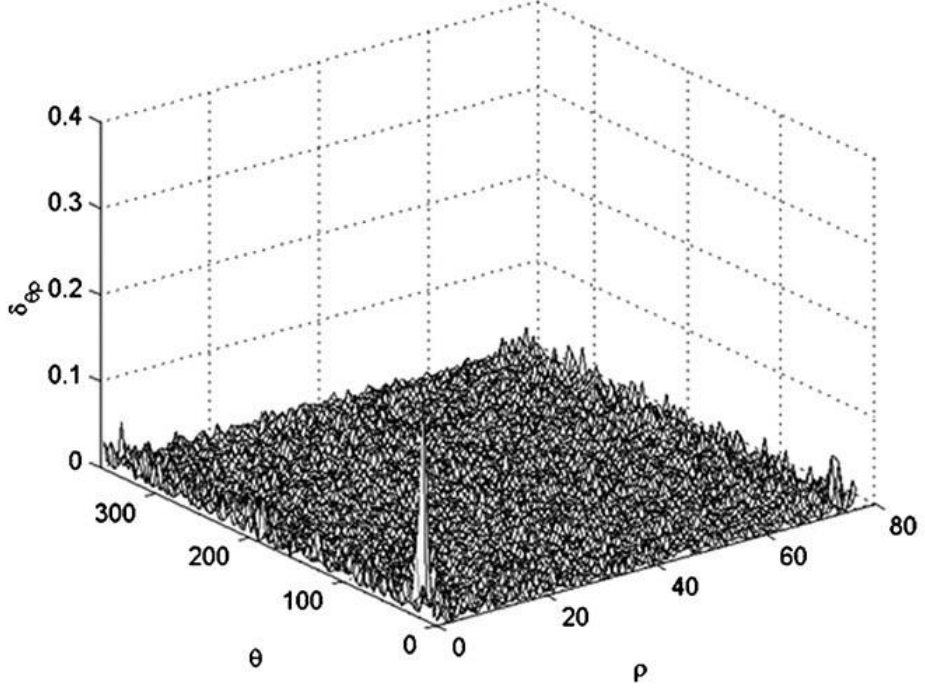


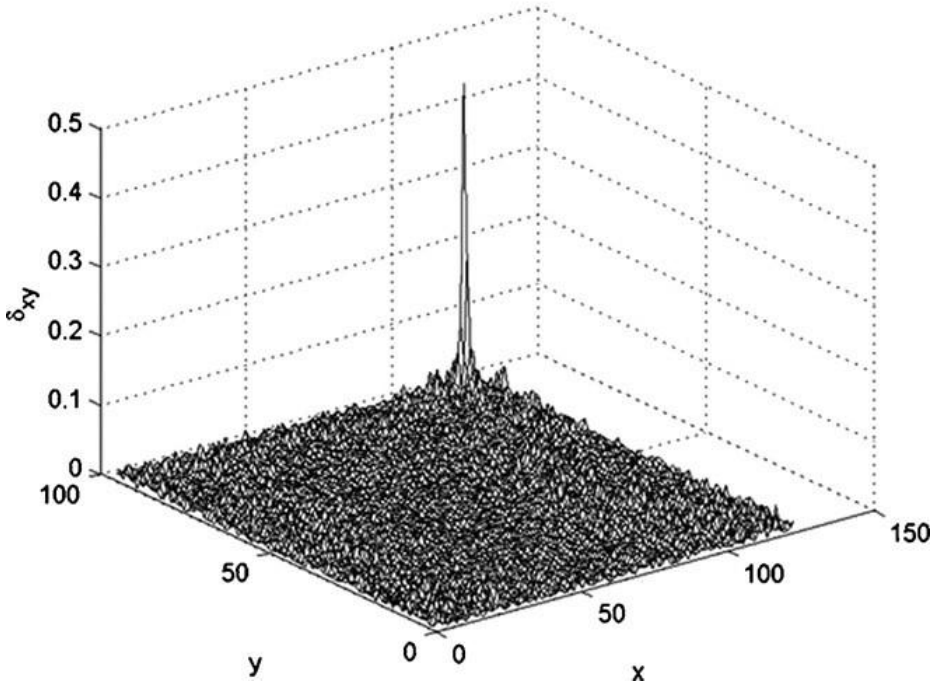
（c）水平方向投影曲线



（d）垂直方向投影曲线

相位相关算法


0.4
0.3
0.2
0.1
0
$\delta_{\theta\rho}$
300
200
100
0
θ
0
20
40
60
80
ρ

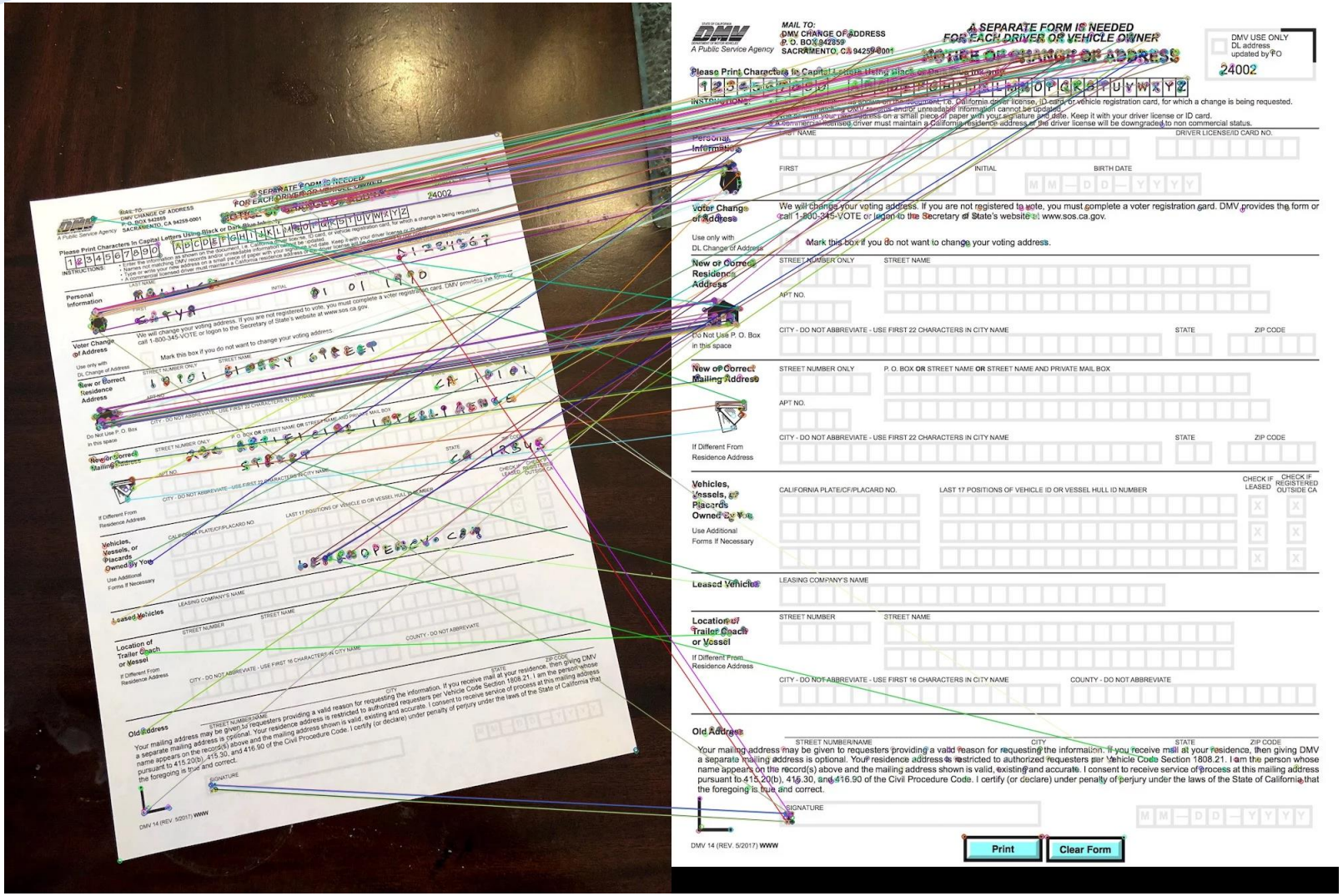

0.5
0.4
0.3
0.2
0.1
0
δ_{xy}
100
50
0
y
0
50
100
150
x

光流法

a

b

基于特征点的方法


MAIL TO:
DMV CHANGE OF ADDRESS
P. O. BOX 942859
SACRAMENTO, CA 94259-0001
A SEPARATE FORM IS NEEDED
FOR EACH DRIVER OR VEHICLE OWNER
NOTICE OF CHANGE OF ADDRESS
DMV USE ONLY
DL address
updated by PO
24002
Please Print Characters In Capital Letters Using Black or Dark Blue Ink Only
INSTRUCTIONS:
Personal Information
LAST NAME
DRIVER LICENSE/ID CARD NO.
FIRST
INITIAL
BIRTH DATE
Voter Change of Address
We will change your voting address. If you are not registered to vote, you must complete a voter registration card. DMV provides the form or call 1-800-345-VOTE or logon to the Secretary of State's website at www.sos.ca.gov.
Use only with DL Change of Address
Mark this box if you do not want to change your voting address.
New or Correct Residence Address
STREET NUMBER ONLY
STREET NAME
APT NO.
Do Not Use P. O. Box in this space
CITY - DO NOT ABBREVIATE - USE FIRST 22 CHARACTERS IN CITY NAME
STATE
ZIP CODE
New or Correct Mailing Address
P. O. BOX OR STREET NAME OR STREET NAME AND PRIVATE MAIL BOX
If Different From Residence Address
Vehicles, Vessels, or Placards Owned By You
CALIFORNIA PLATE/CF/PLACARD NO.
LAST 17 POSITIONS OF VEHICLE ID OR VESSEL HULL ID NUMBER
CHECK IF LEASED
CHECK IF REGISTERED OUTSIDE CA
Use Additional Forms If Necessary
Leased Vehicles
LEASING COMPANY'S NAME
Location of Trailer Coach or Vessel
STREET NUMBER
COUNTY - DO NOT ABBREVIATE
CITY - DO NOT ABBREVIATE - USE FIRST 16 CHARACTERS IN CITY NAME
Old Address
STREET NUMBER/NAME
CITY
STATE
ZIP CODE
Your mailing address may be given to requesters providing a valid reason for requesting the information. If you receive mail at your residence, then giving DMV a separate mailing address is optional. Your residence address is restricted to authorized requesters per Vehicle Code Section 1808.21. I am the person whose name appears on the record(s) above and the mailing address shown is valid, existing and accurate. I consent to receive service of process at this mailing address pursuant to 415.20(b), 415.30, and 416.90 of the Civil Procedure Code. I certify (or declare) under penalty of perjury under the laws of the State of California that the foregoing is true and correct.
SIGNATURE
DMV 14 (REV. 5/2017) WWW
Print
Clear Form

基于硬件：陀螺仪

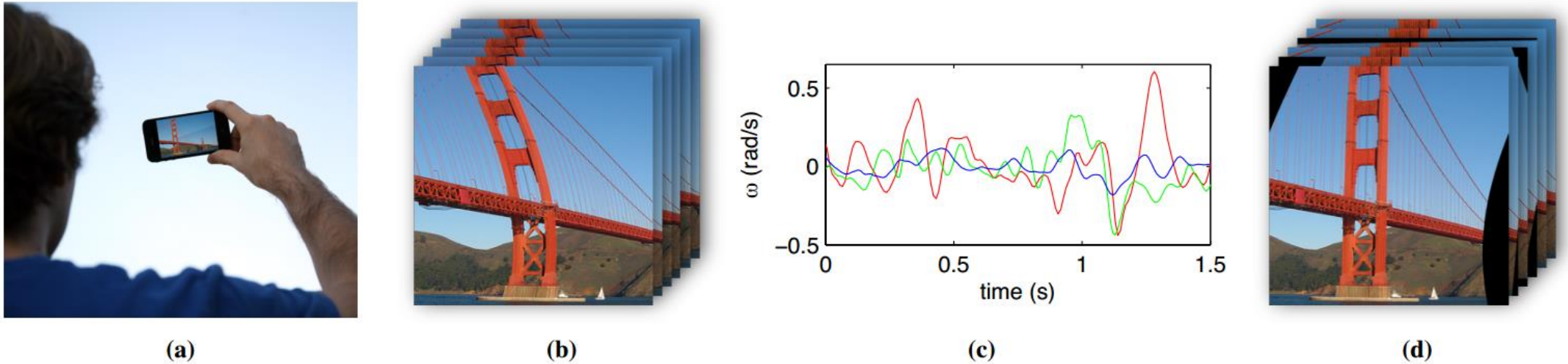
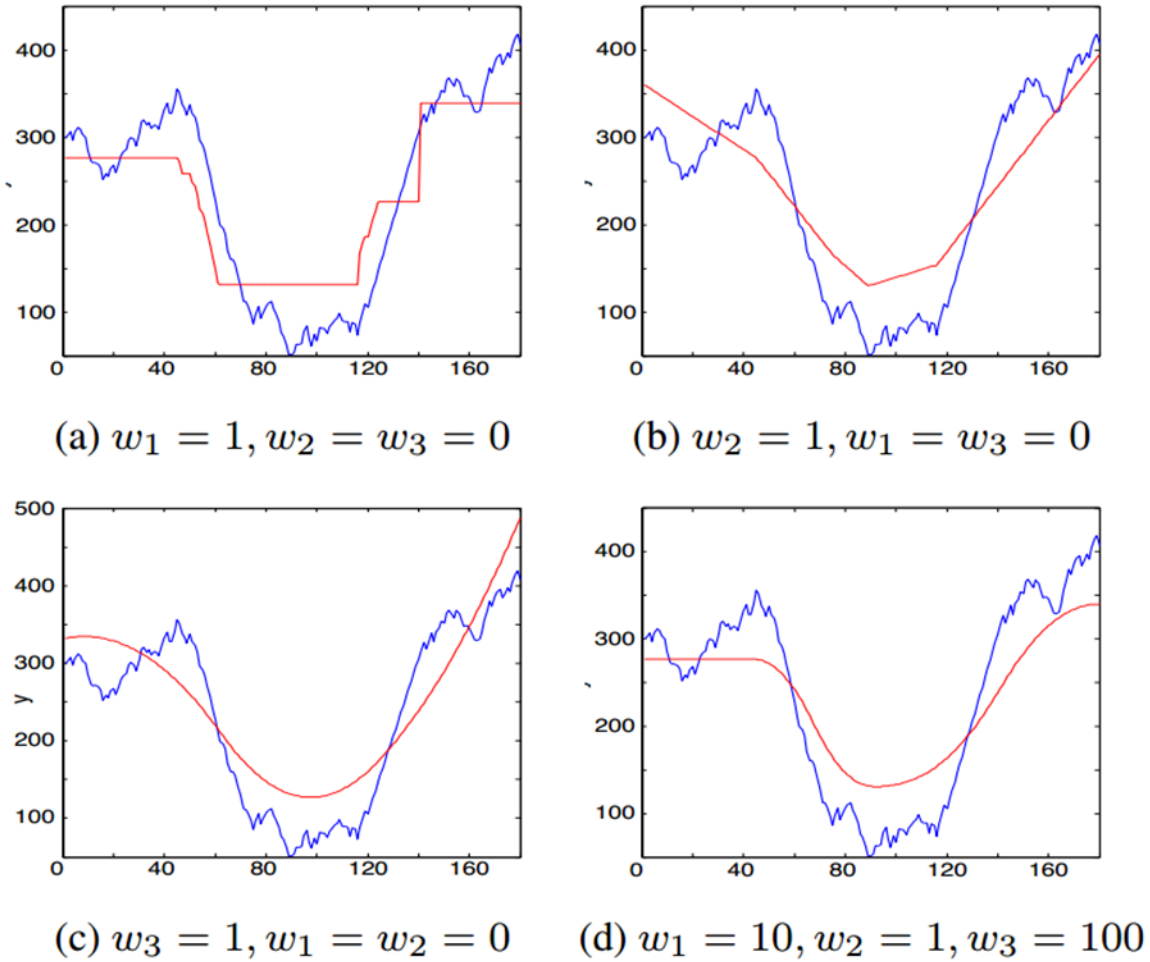


Figure 1: *(a) Videos captured with a cell-phone camera tend to be shaky due to the device's size and weight. (b) The rolling shutter used by sensors in these cameras also produces warping in the output frames (we have exagerrated the effect for illustrative purposes). (c) We use gyroscopes to measure the camera's rotations during video capture. (d) We use the measured camera motion to stabilize the video and to rectify the rolling shutter. (Golden Gate photo courtesy of Salim Virji.)*

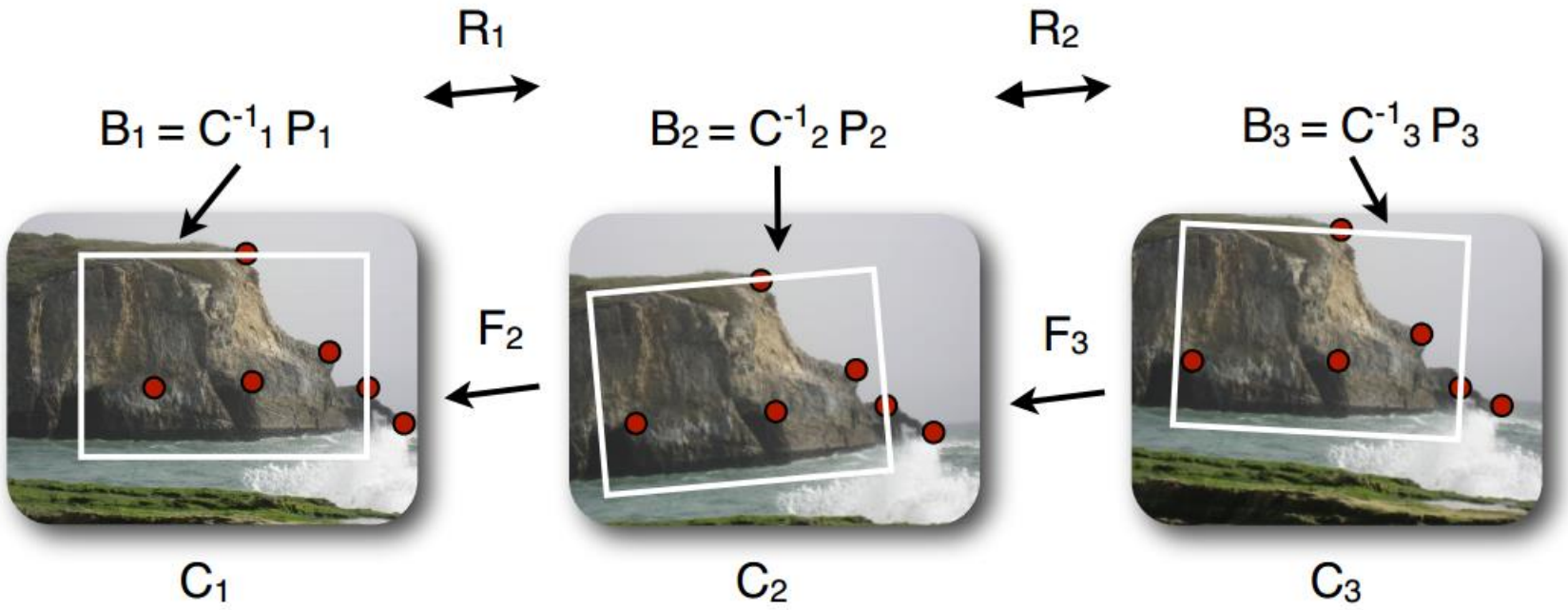
2.计算新的平滑的摄像机路径（运动补偿）

- 运动路径平滑
- 运动路径拟合



(a) $w_1 = 1, w_2 = w_3 = 0$ (b) $w_2 = 1, w_1 = w_3 = 0$

(c) $w_3 = 1, w_1 = w_2 = 0$ (d) $w_1 = 10, w_2 = 1, w_3 = 100$

3.基于平滑路径合成新的稳定的视频（图像修补）

- 剪切处理
- 填补空白边缘